Phase 3 Report

Early Detection of AD/ADRD: A Robust, Fair, and Explainable Framework for Cognitive Score Prediction

Authors: Suhas D. Parandekar[1], Ekaterina Melianova[2], Artem Volgin[3]

Overview

Despite significant advances in machine learning for large-scale data, applying these techniques to smaller, tabular datasets - common in real-world settings - remains challenging due to complex feature interactions, data heterogeneity, and subgroup-specific patterns. In Phase 3 of this competition, we refined our Phase 2 model to develop a lightweight and flexible approach capable of working with similar datasets, aiming to produce solutions that are generalizable, robust, and fair.

For this study, we analyzed harmonized datasets covering four countries: Mexico (MHAS and Mex-Cog), India (LASI and LASI-DAD), the UK (ELSA and ELSA-HCAP), and the US (HRS and HRS-HCAP). As our target, we used the harmonised general cognitive function score from the Harmonized Cognitive Assessment Protocol (HCAP). Unlike Phase 2, the outcome is now measured in standard deviation units, reflecting relative cognitive functioning across individuals. These harmonised scores are derived through confirmatory factor analysis and provide a reliable, cross-nationally comparable measure of cognitive health (Gross et al. 2023).

Because surveys labeled waves inconsistently, we harmonized variables across countries by renumbering waves to align timelines, retaining up to five waves prior to the cognitive assessment. The US and UK datasets included all five waves, Mexico retained four, and India only one. Missing waves were represented as missing values, enabling flexible merging of countries with different survey histories. These datasets include rich information on social determinants of health, demographics, health conditions, and functional status, offering a comprehensive view of factors influencing cognitive health in older adults.

Our pipeline prepared data for modeling by converting ordinal variables to numeric formats, one-hot encoding categorical variables, and applying log transformations to numerical features. It created new indicators for age groups and numeric country codes for consistent grouping. Group-level statistics (e.g., means, variance, and distributional features) were computed for combinations of country and age group and merged back into individual records as additional predictors. We also used flag variables to indicate the presence of the respondent in each wave.

For predicting the cognitive score, we used three boosting models—LightGBM, CatBoost, and XGBoost—which were trained on the processed dataset prepared as described above. Each model underwent hyperparameter tuning via cross-validation and Optuna to minimize prediction error. After tuning, the models were fit to the full data, optionally applying inverse-frequency weights to improve fairness across subgroups. Predictions included SHAP values for feature attribution, and individual prediction intervals were estimated using Jackknife+ to quantify uncertainty. Finally, an ensemble integrated predictions from all three models, resulting in a pipeline that combined diverse algorithms, fairness adjustments, and interpretability measures to enhance generalisability, robustness, and practical utility.

[1] Senior Economist at The World Bank, US
[2] PhD graduate at the University of Bristol, UK
[3] PhD graduate at the University of Manchester, UK

Performance

Model performance across groups and datasets

The model's predictive accuracy was assessed using RMSE on a held-out test set comprising 20% of observations, yielding an overall RMSE of 0.676. Performance varied across subgroups: it was best for participants in the US (RMSE = 0.615) and lowest for those in the UK (RMSE = 0.773), likely due to a smaller UK sample size (n = 266) and potential regional differences. Gender differences were modest, with RMSEs of 0.689 for males and 0.666 for females, while predictions were less accurate for individuals with only primary education (RMSE = 0.698) than for those with tertiary education (RMSE = 0.547), suggesting challenges in predicting outcomes for less educated groups (see Table 1).

Table 1. RMSE by countries, gender, and education

	Country				**Gender**		**Education**		
	India	Mexico	UK	US	Male	Female	Primary	Secondary	Tertiary
RMSE	0.689	0.677	0.773	0.615	0.689	0.666	0.698	0.682	0.547
N	819	407	266	660	944	1208	1180	688	258

Fairness assessment and opportunities for improvement

To quantify disparities in prediction errors across subgroups, we introduced a Fairness Index (FI), $FI = 1 - \frac{max\ (RMSE_{subgroup})}{Range\ of\ RMSE}$, where $RMSE_{subgroup}$ is the RMSE calculated for specific subgroups within a dataset, such as women or age 80+. An FI of 1 indicates perfect fairness, where all subgroups have equal RMSE (range=0), while an FI of 0 reflects maximum unfairness, with the minimum RMSE being 0 (range=max RMSE).

Prior to any adjustments, the model's FI was 0.795 for country, 0.967 for gender, and 0.783 for education, suggesting moderate disparities, especially by country and education. Applying fairness constraints improved the FI to 0.823, 0.997, and 0.875, respectively, but slightly increased the overall RMSE (= 0.676), as shown in Table 2. This underscores the trade-off between reducing subgroup disparities and preserving overall accuracy.

Table 2. Fairness metrics with and without correction

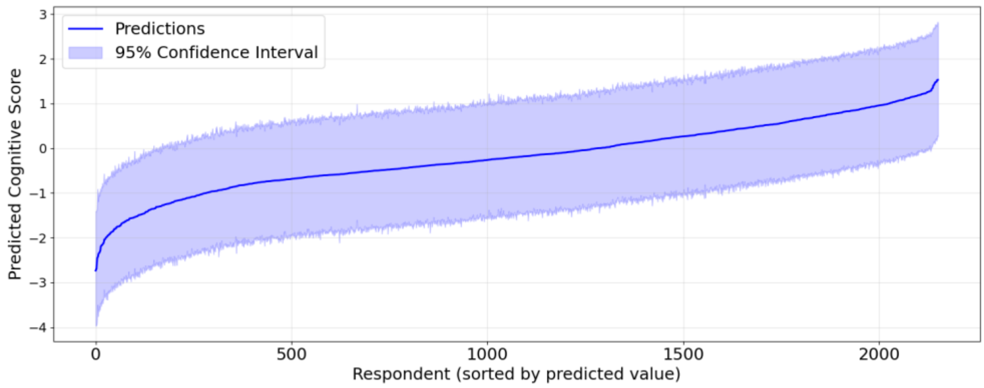
	Main Index	**Fair Index**	**Fair RMSE**
Country	0.795	0.823	0.717
Gender	0.967	0.997	0.740
Education	0.783	0.875	0.762

Additional reduction in performance disparities for these groups may require collecting more data in underrepresented populations, incorporating additional features that capture socioeconomic or regional factors, or developing subgroup-specific models.

Clinical use

An important strength of our approach is its ability to generate confidence intervals for each individual prediction using the **Jackknife+** method (Barber et al., 2020) These intervals provide valuable insight into prediction uncertainty and can support more informed and cautious clinical decision-making. The figure below shows predictions sorted by their estimated values, along with the corresponding 95% confidence intervals, illustrating how uncertainty varies across the range of predictions. While Jackknife+ offers valuable uncertainty estimates, these intervals depend on representative data and may miss important factors, especially in unique subgroups, which clinicians should consider when interpreting results.

Figure 1. Individual predictions with 95% CI

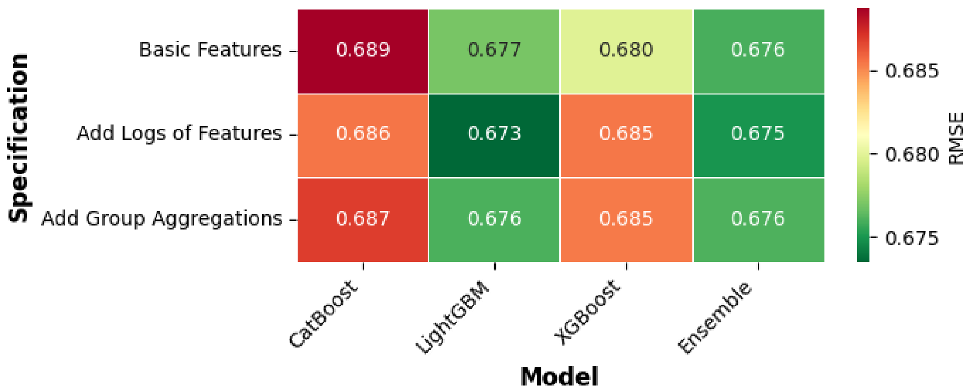


Methodology

As stated in the overview, we evaluated three popular gradient boosting models - CatBoost, LightGBM, and XGBoost - and an ensemble combining their predictions. To test whether additional feature engineering could improve performance, we explored adding logarithmic transformations of numeric variables and features aggregated over key socio-demographic variables - as in our Phase 2 report.

As shown in figure below, the ensemble consistently achieved the best performance, reaching an RMSE of 0.676 with basic features. Although introducing logs and group aggregations led to slight improvements in some individual models, these changes did not significantly reduce the ensemble's error. The differences across feature specifications were modest, suggesting the ensemble already captured most of the available predictive signal. Given the limited performance gains from added complexity, and considering our aim for interpretability, we selected the ensemble of the three boosting models with basic features as our final solution.

Figure 2. RMSE by Model and Specification



Data

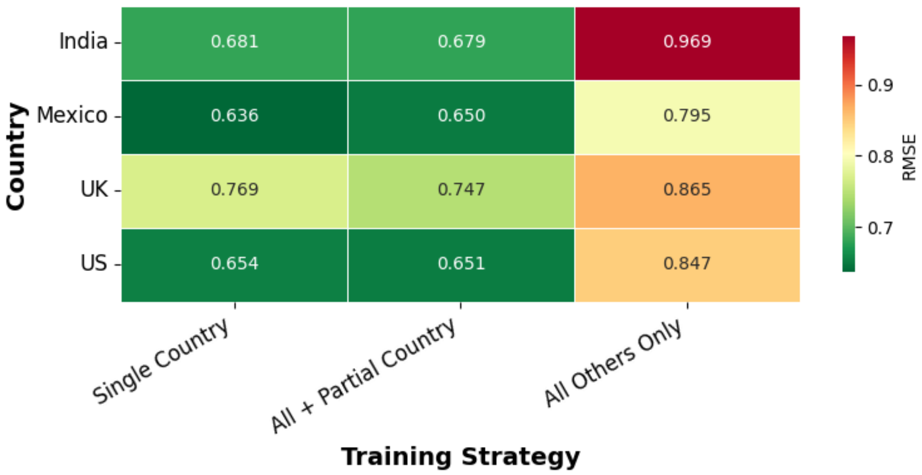
The **generalizability** of our model was assessed across *countries, survey waves,* and *scenarios with fewer observations or features* to understand how existing datasets contribute to performance.

Generalisability across countries

To examine cross-country learning, we compared models trained and tested within a single country against models trained on pooled data from all four countries but tested on one country. For most countries, except Mexico, adding data from other countries reduced RMSE, indicating that shared information improves predictions across settings.

We also tested the model's ability to generalize to entirely unseen countries by training on three countries and testing on the fourth. While this resulted in a roughly 10-30% increase in RMSE compared to partial holdout, performance remained acceptable, suggesting some capacity for transfer learning to new populations.

Figure 3. RMSE by Country Training Strategies



Generalisability across survey waves

Additionally, we examined how removing recent survey waves impacts the model's ability to make long-term predictions. Excluding India, which has only a single wave, we found that omitting the most recent wave increased RMSE from a baseline of 0.663 to 0.709. Removing two and then three recent waves further raised RMSE to 0.735 and 0.744, respectively. These results indicate that recent data improve predictive accuracy for future outcomes. However, the decline in accuracy was not severe, suggesting that even with older data, the model can still produce valid predictions extending five years or more into the future.

Generalisability under reduced data

We also tested our model's robustness by randomly removing up to half of the features or observations. Encouragingly, RMSE remained largely stable, demonstrating that the model is resilient even with substantial reductions in data size or dimensionality.

Table 3. Generalizability results based on removing features and observations

	Features	Observations
Removed 10%	0.685	0.663
Removed 30%	0.689	0.675
Removed 50%	0.690	0.682
Removed 70%	0.770	0.677
Removed 90%	0.834	0.705

To improve performance and generalizability, future data collection should focus on expanding sample sizes in underrepresented countries, gathering additional survey waves for long-term prediction, and incorporating new variables capturing socioeconomic, regional, and clinical factors.

Contributing factors

For each feature, its SHAP values - which represent how much it contributes to the model's predictions - were statistically regressed on the feature's original values to estimate how changes in the feature relate, on average, to changes in predicted cognitive scores. The resulting coefficients indicate both the strength and direction of these relationships, while confidence intervals help assess their reliability. This transforms local SHAP explanations into a global understanding of feature importance and effect direction across the study population (Molnar 2020).

Table 4. TOP-10 Positive Coefficients

Wave	Feature	Coefficient	95% CI
-	Harmonized education level	0.484	[0.478,0.491]
4	People act as if the respondent is not smart: [less than once a year]	0.013	[0.013,0.013]
5	Whether currently receiving private pension payments [yes]	0.011	[0.01, 0.011]
3	Self-reported marital status [married]	0.011	[0.01, 0.011]
5	Whether self-employed [self-employed]	0.009	[0.009,0.009]
5	Cantril ladder rating (10-point scale)	0.009	[0.008, 0.01]
4	Treated with less courtesy or respect [less than once a year]	0.008	[0.008,0.009]
5	Mother Alive [Yes}	0.008	[0.008,0.008]
5	Received poorer service from doctors in daily life [a few times a year]	0.008	[0.008,0.008]
5	Whether unemployed [yes]	0.006	[0.006,0.006]

Table 5. TOP-10 Negative Coefficients

Wave	Feature	Coefficient	95% CI
5	Lives in rural or urban [rural]	-0.111	[-0.113,-0.109]
4	Lives in rural or urban [rural]	-0.021	[-0.022,-0.021]
4	Respondent lack of friends support summary mean score (7-items)	-0.021	[-0.023,-0.019]
2	Loneliness summary mean score (4-items)	-0.015	[-0.016,-0.014]
5	Self-reported marital status [partnered]	-0.014	[-0.015,-0.014]
1	Respondent lack of friends support summary mean score (7-items)	-0.013	[-0.015,-0.012]
4	Received poorer service than other people [at least once a week]	-0.012	[-0.012,-0.011]
1	Loneliness summary mean score (4-items)	-0.011	[-0.012,-0.011]
3	Respondent lack of friends support summary mean score (7-items)	-0.01	[-0.011, -0.01]
2	Respondent lack of friends support summary mean score (7-items)	-0.01	[-0.011,-0.009]

The results show that higher education is the strongest positive predictor of cognitive score, supporting evidence that educational attainment builds cognitive reserve and may delay Alzheimer's onset (Stern, 2012; Livingston et al., 2020). Positive links with marital status, private pensions, self-employment, and life satisfaction suggest that social integration and financial security promote better cognition (Zhang et al., 2015). Conversely, rural residence, loneliness, lack of social support, and experiences of discrimination are strong negative predictors, reflecting research on social isolation, rural health disparities, and psychosocial stress as dementia risks (Kuiper et al., 2015; Saenz et al., 2022; Hudomiet et al., 2024). Overall, the model highlights how education, social connections, and perceived fairness shape cognitive aging trajectories.

Future directions

To advance early prediction of AD/ADRD, future work should prioritize fairness-aware machine learning methods to reduce subgroup disparities, further explore cross-dataset transfer learning to improve performance in underrepresented populations, and collect additional longitudinal data to enhance long-term forecasts. Integrating new data sources - such as neuroimaging, genetic markers, digital biomarkers, and detailed social determinants - will also be critical for capturing the full range of factors influencing disease risk. Despite current challenges, our modular pipeline offers a foundation for applying these methods broadly and guiding future research toward more robust, equitable early detection of AD/ADRD.

To support further progress, we have developed an automated pipeline designed for reproducibility and extensibility. It allows efficient application of our methods to other datasets, thorough assessment of errors and biases, exploration of social determinants of health through SHAP values computation and person-level uncertainty evaluation. We believe this tool will facilitate ongoing research and adaptation of predictive models to new contexts and populations.

References

Barber, R. F., Candès, E. J., Ramdas, A., & Tibshirani, R. J. (2020). Predictive inference with the jackknife+. arXiv preprint arXiv:1905.02928. https://doi.org/10.48550/arXiv.1905.02928

Gross, A. L., Li, C., Briceño, E. M., Rentería, M. A., Jones, R. N., Langa, K. M., ... & Kobayashi, L. C. (2023). Harmonisation of later-life cognitive function across national contexts: results from the Harmonized Cognitive Assessment Protocols. The lancet Healthy longevity, 4(10), e573-e583.

Molnar, C. (2020). Interpretable machine learning. Lulu. com.

Stern, Y. (2012). Cognitive reserve in ageing and Alzheimer's disease. The Lancet Neurology, 11(11), 1006-1012.

Livingston, G., et al. (2020). Dementia prevention, intervention, and care: 2020 report. *Lancet*, 396(10248), 413-446.

Zhang, M., Gale, S. D., Erickson, L. D., Brown, B. L., Woody, P., & Hedges, D. W. (2015). Cognitive function in older adults according to current socioeconomic status. Aging, Neuropsychology, and Cognition, 22(5), 534-543.

J. L. Saenz, B. Downer, M. A. Garcia, and R. Wong, 'Rural/urban dwelling across the life-course and late-life cognitive ability in Mexico', *SSM - Popul. Health*, vol. 17, p. 101031, Mar. 2022, doi: 10.1016/j.ssmph.2022.101031.

Kuiper, J. S., et al. (2015). Social relationships and risk of dementia: A systematic review and meta-analysis. *Ageing Research Reviews*, 22, 39-57.

P. Hudomiet, M. D. Hurd, and S. Rohwedder, 'Identifying Early Predictors of Cognitive Impairment and Dementia in a Large Nationally Representative U.S. Sample', RAND Corporation, Dec. 2024.